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Research paper

Enhancing anomaly detection in electrical consumption profiles through computational intelligence

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ABSTRACT

The advancement of society and the raising of people's standards of living depend heavily on electricity in today's world. The "zero energy buildings" idea, which recommends that buildings become self-sufficient in renewable energy to prevent the emission of CO2 into the environment, is now one of the most significant initiatives connected to building energy efficiency. This article describes a computational intelligence method to detect anomalous variations in a facility's energy use and infer a potential cause of such changes. The model is built using five sets of historical power consumption data from three buildings spread across four nations (Ecuador, Spain, France, and Canada), which are categorized based on the anomaly type each piece of data represents. Through a statistical study of the confidence interval, the proposed method, first determines the consumption patterns for each day of the week in each of the building's data sets. After normalizing the day to be studied toward its "Z" value, it is then cataloged using a machine learning model. The proposed method is evaluated in comparison to a purely statistical method called SAEEC methodology and it is discovered that the proposed method offers a relative improvement in accuracy, false positive rate (FPR), and false negative rate (FNR) of 12.41%, 42, 36%, and 42.45%, respectively, for the detection of atypical values in electrical energy consumption.

1. Introduction

Maximizing resources and adopting energy efficiency policies and systems have become increasingly crucial in the realm of building operations, planning, and energy management. Recent advancements in anomaly detection using computational intelligence have emerged as significant contributors to this field. Notably, the application of Hyperdimensional Computing (HDC) in smart grids and deep learning methodologies for anomaly detection in electrical consumption data have provided new insights and tools for energy management and the identification of anomalous consumption patterns (Wang et al., 2022; Serrano-Guerrero, 2020). According to recent studies, the analysis of patterns in the use of electrical power enables the observation of how energy is demanded within a building, thus making it easier to apply strategies with the goal of enhancing energy efficiency. Within this context, the evaluation of fluctuations in consumption profiles enables the identification of potential correlations with various events (Serrano-Guerrero et al., 2020). Recent research has expanded our understanding of Electrical Consumption Patterns (ECPs) by incorporating advanced methodologies such as Long Short-Term Memory (LSTM) neural networks and machine learning techniques. These state-of-the-art approaches have shown significant potential in detecting irregularities in electricity consumption, thereby offering a more nuanced and effective means of analyzing ECPs for energy efficiency improvements (Wang et al., 2019).

Understanding the present and future energy behavior of a building can prevent issues related to energy supply or unnecessary energy allocation in the medium and long term (Serrano-Guerrero, 2020). An electricity consumption profile, also known as a daily load profile (DLP), represents the daily energy demand patterns exhibited by consumers (Serrano-Guerrero, 2020). Analyzing these patterns enables the implementation of strategies aimed at enhancing efficiency (Serrano-Guerrero, 2020). Within the literature, methods that consider daily behavior are referred to as consumption profile methods (Kardi et al.,

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2021).

Examining the DLP provides insights into electrical consumption patterns (ECPs), which offer observations regarding energy demand behaviors within a facility (Serrano-Guerrero et al., 2018; Romero and Serpa-Andrade, 2021). This information can be utilized to generate alerts, reduce maintenance expenses, and respond promptly to any instances of unusual energy consumption (Alfares and Nazeeruddin, 2002). The primary goal of ECPs is to improve energy efficiency, which holds economic, environmental, and social benefits (Serrano-Guerrero, 2020).

Certain researchers, such as Seem (2005a, 2007) and Li et al. (2010), focus solely on average daily energy consumption and maximum daily usage for pattern recognition, making it unfeasible to conduct a comprehensive examination of ECPs (Capozzoli et al., 2018). The establishment of energy time series within predetermined time intervals creates a barrier for conducting analysis in real-time or near-real-time. In contrast, alternative research studies referenced as (Jokar et al., 2016) utilize support vector machines and unsupervised learning algorithms such as k-means, while Fenza et al. (2019) employ LSTM and Statistical neural networks to detect irregularities in electricity consumption. Both approaches require significant effort when it comes to selecting appropriate training data, and there is still a need for further exploration in the field of time series analysis.

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Detecting contextual anomalies has become a recent and relatively unexplored challenge in the field of power monitoring. Recent developments have addressed specific challenges associated with the high-dimensional nature of data and the handling of mixed-type data, which are crucial for effective anomaly detection in electrical consumption profiles. Innovative approaches have been devised to manage the complexities inherent in these data types, enhancing the accuracy and robustness of anomaly detection systems (Oprea and Bâra, 2021; Himeur et al., 2021). In their research, Araya et al. (2017) introduce an ensemble system for anomaly detection that incorporates majority voting to combine various classifiers for identifying outliers. Similarly, Hayes and Capretz (2015) propose a contextual anomaly detection approach that also relies on majority voting. However, these methods fail to take into account the structural characteristics of time series data and demand significant computational resources. Furthermore, in the data pre-processing stage, only negative values were deemed noisy in Hayes and Capretz (2015) Conversely, Cui and Wang (2017) introduce a polynomial regression model and Gaussian distribution-based method for anomaly detection. Although this study achieved a zero false negative rate (FNR) in all tested cases, it only identifies contextual anomalies using a higher threshold, which means that low consumption outliers cannot be detected.

In a different study, Fan et al. (2018) conducted a distinct study where they examined diverse architectures and training techniques for autoencoders in order to identify anomalies in building energy data. This research applies statistical techniques to handle large-scale, high-dimensional data environments, but it reveals scalability and suitability challenges. Each stage of the process requires significant computational efforts, this involves the identification of influential factors in energy consumption, such as the month, day, and hour, as well as determining the widely acknowledged dominant period of 24 h. To enhance the learning capabilities of autoencoders, different levels of masking noise are introduced to create corrupted subsequences.

Researchers have conducted extensive studies on Electrical Consumption Patterns (ECPs) across five principal application domains: (i) prediction of electrical usage, (ii) management and optimization of energy consumption, (iii) forecasting electricity prices, (iv) categorization of consumption patterns, and (v) identification of unusual consumption patterns (Cai et al., 2019). These studies are pivotal in various fields, including electrical and economic sector planning, energy efficiency initiatives, anomaly detection, tariff establishment, cost reduction strategies, fault identification, and related areas. To achieve these objectives, each methodology employs a diverse array of tools and techniques. Nevertheless, certain weaknesses have been identified across these domains, which can be categorized into four main areas:

a) **Comprehensive Analysis Deficit**: A significant shortcoming is the lack of comprehensive consideration of the Electrical Consumption Pattern (ECP), which hinders a holistic understanding of consumption behaviors.
b) **Time-Series Data Structure Analysis**: There is a notable need for improvement in techniques concerning the analysis of the time-series data structure. This gap limits the depth of insights that can be drawn from the data.
c) **Training Data Selection Challenges**: The process of selecting appropriate training data for models is time-consuming and labor-intensive, posing a considerable challenge in model development.
d) **Generalization Limitation**: The reliance on singular case studies in many research efforts leads to a generalization challenge, as findings may not be applicable to broader or varying contexts.

There are works in the field of electricity prediction that have shown promise, but they do not consider patterns of electrical consumption (Weakness type "a") and only one case study is examined (Weakness type "d"). These works use techniques like Artificial Neural (Serrano-Guerrero et al., 2017; Escrivá-Escrivá et al., 2011), Support Vector Machines (Jain et al., 2014), and Data Mining with Bayesian networks (Singh and Yassine, 2018).

Hierarchical Clustering (Jota et al., 2011), Adaptive Neural Fuzzy Inference System combined with Heuristic Optimization (Shareef et al., 2018), have all been used in the domain of management and optimization of energy consumption and have shown promise, however these studies also do not include consumption patterns. Only one case study is examined (weakness type "a") (Weakness type "d").

The best results in the field of forecasting of electricity prices come from studies using recursive filters, seasonal models, and univariate AR and ARMA models (Janczura et al., 2013) (Cuaresma et al., 2004). However, in this field, in addition to the weaknesses of types "a and d," there is also the weakness of type "b," which refers to studies that do not account for the time structures of the data series.

Statistics, hierarchical clustering (Seem, 2005b), data mining, PSO k-means, and machines of support vectors (Cai et al., 2019) are techniques used for categorization the type of days of electricity consumption. Furthermore, recent advancements in machine learning techniques, such as frequent pattern analysis and the application of sophisticated classification algorithms, have proven to be highly effective in managing large datasets and identifying anomalous patterns in electricity consumption. These methods offer enhanced precision and scalability in the analysis of electrical consumption patterns, addressing some of the limitations of traditional statistical approaches (Oprea and Bâra, 2021). These techniques have results that demonstrate relatively high accuracy compared to other methods and techniques, however, these works do not take into account the patterns of electrical consumption (Weakness type "a").

The best works in the field of Identification of unusual patterns (Outliers) employ methods like analysis of canonical variables (Li et al., 2009), K-means, and support vector machines (Milton et al., 2018), but they do not account for patterns of electrical consumption (Weakness type "a"), time structures of the data series (Weakness type "b"), and only one case study (Weakness type "d") is examined.

Finally, in the field of outlier detection, methods such as Statistical Analysis and Hierarchical Clustering (Seem, 2007), as well as other techniques such as Symbolic Data Analysis (Li et al., 2009), ensemble Method Based on Autoencoder with Unsupervised Learning (Fan et al., 2018), and C-means based on clustering diffuse (Angelos et al., 2011) present satisfactory results but do not account for the patterns of electrical consumption (Weakness type "a"). However, the world of Neural networks and LSTM statistics (Fenza et al., 2019) does account for these patterns. Other works include the ensemble anomaly detection framework and overlapping sliding windows (Araya et al., 2017), support for vector machines and k-means (Jokar et al., 2015), multivariate clustering majority vote classifier (Hayes and Capretz, 2015), gaussian distribution and polynomial regression model (Cui and Wang, 2017), If they take into consideration the consumption pattern and extrapolate for various case studies, but do not account for the time structures of the data series (type "b" weakness), and the data processing for the system's training phase demands a lot of labor.

In a previous study, the authors introduced the methodology termed Seasonality Analysis of Electricity Consumption (SAECC), which is grounded in the definition of Electrical Consumption Patterns (ECPs). This methodology focuses on assessing changes in the Daily Load Profiles (DLP) of buildings, employing a purely statistical approach as detailed in (Serrano-Guerrero et al., 2020). The SAECC method facilitates the cataloging of anomalous electricity consumption through a multi-criteria system. However, one of the critical challenges of this approach lies in its implementation: configuring, training, and modeling a system to identify irregularities in electrical consumption are processes that demand considerable time and specialized expertise in electrical energy systems.

Building upon these foundations, the present study introduces a novel machine learning-based method for detecting and cataloging changes in an installation's DLP. This method not only aligns with, but also advances beyond, recent advancements in computational intelligence. It specifically addresses complex challenges such as handling high-dimensional data, which often renders traditional analytical methods inadequate, and effectively identifying anomalies in a landscape where standard detection algorithms may fail. Our approach leverages sophisticated machine learning techniques to markedly enhance the accuracy and efficiency of anomaly detection in electrical consumption profiles, thereby offering robust solutions to some of the most critical issues previously identified in the field.

By exploiting the capacity of artificial intelligence to generalize behaviors (models) from information (data) provided as examples, our work transcends the limitations of conventional methodologies. This novel approach is rooted in a deep understanding of the intricate patterns of electrical consumption, enabling a nuanced detection of anomalies that were previously undetectable with existing methods. Consequently, it holds the potential to significantly enhance electricity management, streamline data analysis procedures, and improve the effectiveness of energy monitoring systems.

The primary contributions of this study are groundbreaking and multifaceted:

- We have developed a unified computational intelligence model capable of discerning and categorizing subtle changes in the daily load profile (DLP) databases of five historical buildings, a feat not previously accomplished in the field. This model leverages precise information, thereby increasing its applicability across various case studies.
- Our Computational intelligence model is designed to be intuitively accessible, not requiring expert knowledge in electrical energy, thus democratizing the process of energy management, maintenance, and planning in building systems.
- We have achieved a significant improvement in the detection of anomalous DLPs, leading to a substantial increase in accuracy and a notable reduction in false positive and false negative rates.
- A key innovation is the identification of contextual anomalies within DLPs, which allows for a more holistic and accurate understanding of energy consumption patterns and irregularities.

The subsequent sections of this paper are organized as follows: Section 2 provides background information, while Section 3 presents a detailed explanation of the methodology. Section 4 discusses the results, and Section 5 provides a comprehensive analysis of the findings. Finally, Section 6 concludes the paper.

2. Background

Electricity consumption, as a time series data, is influenced by a stochastic discrete process. When examining time series data, it is evident that it encompasses four key components, representing a series of statistical observations acquired over a specified time span: irregular, cyclical, seasonal, and trend components (Lind et al., 2008). Fluctuations observed in this data series arise from a variety of factors, such as changes in ambient temperature, alterations in work schedules, and vacation periods (Serrano-Guerrero et al., 2020). Despite the inherent unpredictability associated with these variables, it is feasible to identify recurring patterns or behaviors in electricity demand (Serrano-Guerrero, 2020).

To model this process, we utilize time series notation where X_t denotes the value of the series at time t. The time series can be formally defined as a set of random variables X associated with a countable time instant t, belonging to the set of natural numbers $\boldsymbol{N}$. This is represented in Eq. 1 as follows:

$$X_t, (t \quad \epsilon \quad \boldsymbol{T}) \quad \text{dado que} \boldsymbol{T} \subseteq \boldsymbol{N} \tag{1}$$

Here, $\boldsymbol{T}$ represents the set of time points at which observations are made, and $\boldsymbol{N}$ denotes the set of natural numbers. This notation clarifies that we are considering a discrete time series, where observations are made at specific instants within a natural time range.

2.1. Anomaly behavior in DLP

Anomalies within a time series data set refer to data points that deviate significantly from the remaining observations (Aggarwal, 2017). In the context of time series, three types of anomalies can be identified:

- **Punctual anomalies** occur when a particular data point displays an unexpected deviation from the rest of the data set (Chandola et al., 2009).
- **Contextual anomalies** arise when data exhibits anomalous behavior in one context while being normal in another context (Chandola et al., 2009).
- **Collective anomalies** are identified when a group of data points collectively deviates from the remaining instances. Although each individual data point may appear typical, the combined existence of these factors signifies an outlier (Chandola et al., 2009).

Detecting outliers in time series data, particularly when dealing with cases involving interdependencies between distinct data samples (contextual anomalies), poses a significant challenge (Aggarwal, 2017). The identification of outliers based on data point ratios becomes particularly intricate in such scenarios.

In a previous study (Serrano-Guerrero et al., 2020), the SAECC methodology was proposed as a statistical approach in order to categorize abnormal energy consumption patterns using a multi-criteria system, a considerable amount of effort is required to establish a comprehensive framework for identifying anomalies in electrical consumption. In order to detect changes in consumption and propose potential causes, the SAECC methodology initially establishes an electrical consumption pattern (ECP) based on a pre-defined database.

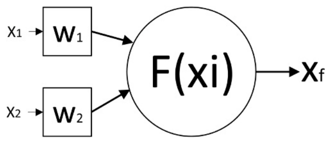

Fig. 1. Diagram of a simple neuron.

Table 1
Testing hypotheses to identify outliers.

		Test	
		Anomalous DLP	**Not Anomalous DLP**
Reality	**Anomalous DLP** (Reject the assumption of the null hypothesis)	True positive (**TP**)	False negative (**FN**)
	Not Anomalous DLP (Accept the assumption of the null hypothesis)	False positive (**FP**)	True negative (**TN**)

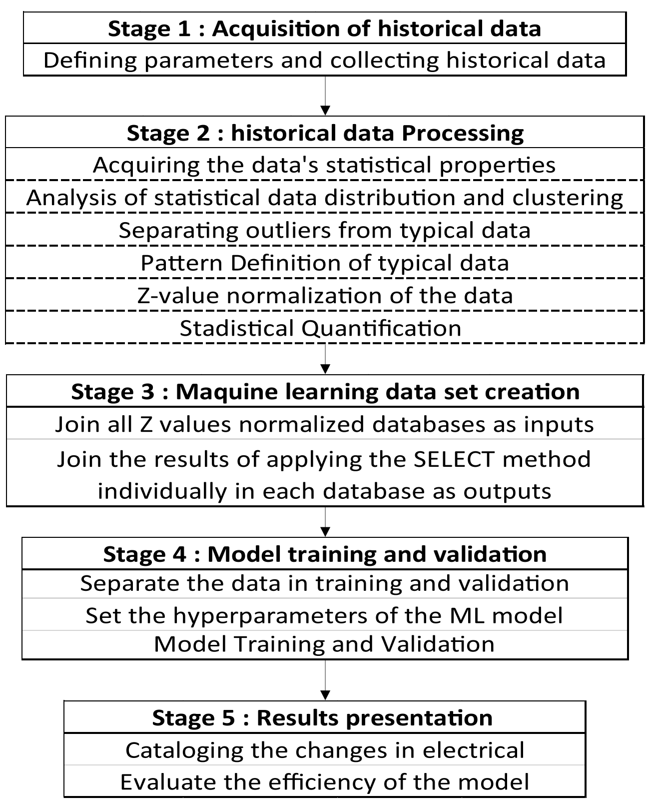

Fig. 2. Methodology proposal.

2.2. Machine Learning

One of the Artificial Intelligence (AI) concepts is the creation of models of concepts from producing simple examples. Based on this fundamental concept emerges the field known as Machine Learning (ML), which represents a domain within the realm of artificial intelligence focused on the development of algorithms and models capable of generalizing behaviors (models) from information (data) given as examples (Contreras Ortiz and Mora Alvarez, 2018). Within the realm of Machine Learning (ML), algorithms can be classified into two primary categories: supervised and unsupervised learning (Contreras Ortiz and Mora Alvarez, 2018). The data or samples in the supervised are labeled either numerically or categorically. The goal of the supervised method is to identify a collection of attributes from the samples that can reliably predict the value of each training set's right label, resulting in a function that outputs a real value in regression problems or a category or class in classification issues. Artificial neural networks are among the machine learning methods that are currently most often employed (Contreras Ortiz and Mora Alvarez, 2018).

2.2.1. Artificial neural networks

In the context of supervised machine learning, artificial neural networks employ a modeling approach that emulates the neural mechanisms of the brain to establish a relationship between a set of input signals and corresponding output variables. This modeling process aims to simulate the cognitive processes involved in information processing and response generation. Different neuron network models have different activation functions, network topologies, and training algorithms (Lantz, 2019).

As can be seen in Fig. 1, each node or "neuron" of the network is made up of a series of inputs (xi), each with a matching weight (wi) that will be optimized during the learning process, and then an output (xf).

The weights of a neural network are commonly initialized randomly, and during the training process, these weights are iteratively adjusted to optimize the network's performance with respect to the specific task at hand (Vasco Carofilis et al., 2020). The performance of the network is evaluated using a loss function, which quantifies the disparity between the expected and actual output generated by the network (Vasco Carofilis et al., 2020).

When building a neural network, it is necessary to define specific predetermined values known as hyperparameters. These hyperparameters encompass the learning rate, the quantity of hidden layers, the number of neurons in each hidden layer, and the selection of activation functions. Optimization of these hyperparameters can lead to improved performance of the neural network.

2.2.2. Hyperparameter optimization

The method of determining the hyperparameter configuration that results in the machine learning model's optimal performance. The procedure is often labor-intensive and laborious. In machine learning, tuning hyperparameters is a time-consuming but essential operation since a model's performance may be influenced by the selection of its hyperparameters. Since manual optimization takes a lot of time, time that may be better spent on other development-related activities, automatic techniques like gradient-powered decision trees are utilized (Vasco Carofilis et al., 2020).

2.2.2.1. Gradient-boosted decision trees. The approach of gradient-boosted decision trees serves as a machine learning methodology that aims to optimize model performance by iteratively adjusting hyperparameters during the learning process (Calzavara et al., 2019). Through a series of iterations, the values of these hyperparameters are fine-tuned to predict the target variable, with the objective of minimizing the loss function that quantifies the disparity between predicted and actual values. The incremental adjustments made at each step, known as gradients, are combined with boosting techniques to expedite the improvement of predictive accuracy towards an optimal level (Calzavara et al., 2019). By leveraging this iterative approach, gradient-boosted decision trees facilitate the effective optimization of hyperparameters, resulting in enhanced model performance and more precise predictions. This methodology allows the model to adapt and refine its hyperparameter settings over time, continually improving its ability to capture complex patterns and relationships within the data. Through the strategic combination of gradient-based adjustments and boosting techniques, the model can effectively navigate the multidimensional space of hyperparameters, finding the optimal configuration that minimizes the loss function and maximizes predictive accuracy (Calzavara et al., 2019). This iterative optimization process contributes to the overall robustness and reliability of the model, enabling it to make

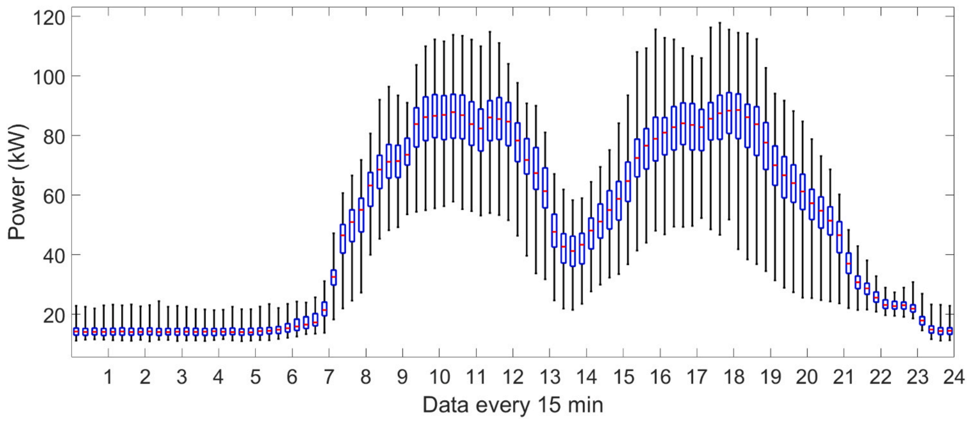

Fig. 3. Electric consumption pattern.

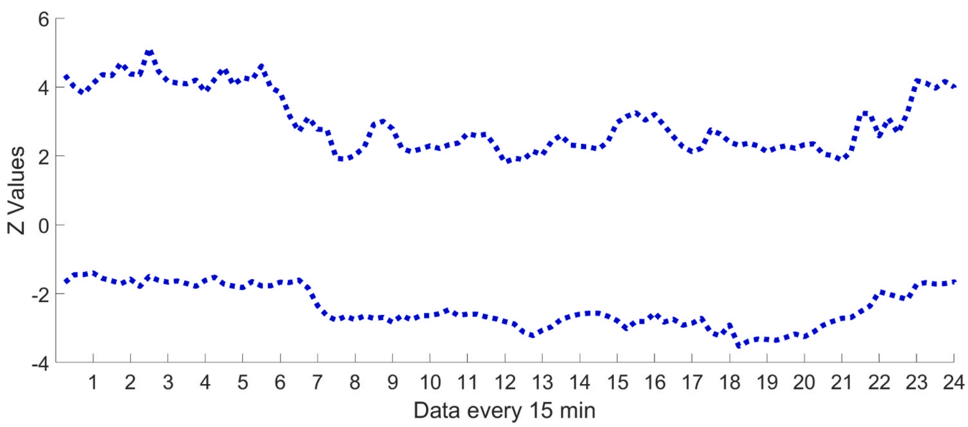

Fig. 4. 0 Electricity consumption pattern in Z Domain.

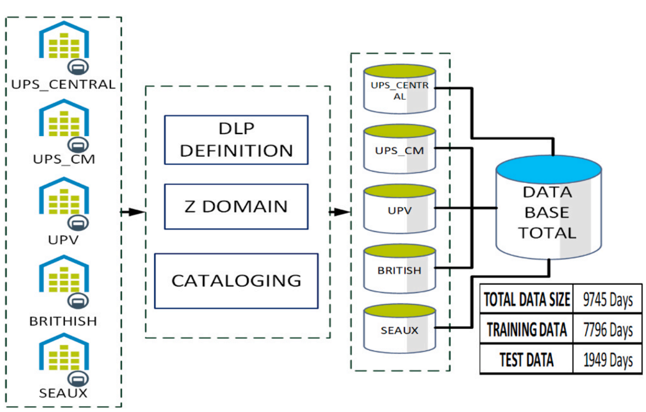

Fig. 5. Data processing and data set creation.

more accurate predictions in various real-world applications.

2.3. Detecting outliers with statistical hypothesis testing

The procedure of hypothesis testing involves the evaluation of a hypothesis's credibility by leveraging probability theory and analyzing statistical data derived from a representative sample (Lind et al., 2012). The hypothesis, serving as a proposition regarding a population, undergoes rigorous scrutiny through statistical analysis to either corroborate or refute its validity. Within this investigative framework, the null hypothesis assumes center stage, drawing particular attention. In the

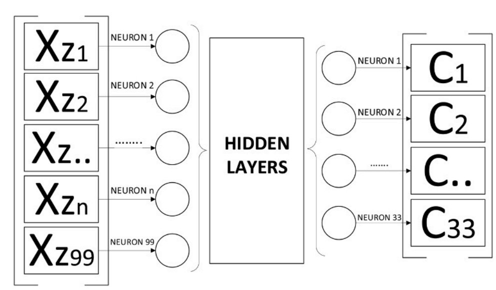

Fig. 6. Proposed neural network architecture.

context of this study, the authors designate the DLP as the null hypothesis, considering it to possess anomalous characteristics. To provide clarity, certain terminologies employed herein are elucidated in Table 1. For instance, false positives (FPs) denote non-anomalous DLPs that are erroneously identified as anomalous.

In hypothesis testing, comparisons are made using evaluation criteria including False Positive Rate (FPR), False Negative Rate (FNR), precision, sensitivity, and specificity. These terms are defined as follows:

- **False Positive Rate (FPR)** is the ratio of the number of false positive cases to the sum of false positive and true negative cases. It is calculated using Eq. 2:

$$FPR = \frac{FP}{FP + TN} \tag{2}$$

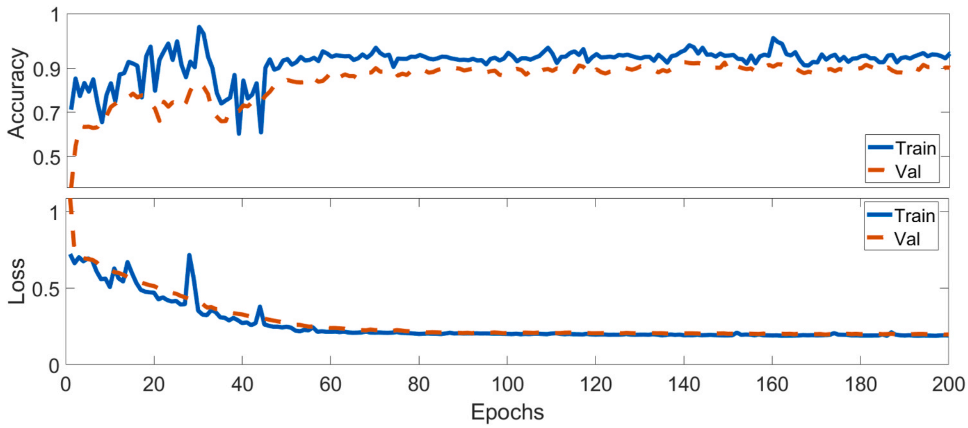

Fig. 7. System Training and Validation.

- **False Negative Rate (FNR)** is the ratio of the number of false negative cases to the sum of false negative and true positive cases. It is given by Eq. 3:

$$FNR = \frac{FN}{FN + TP} \tag{3}$$

- **Precision** refers to the ratio of correctly identified outliers among the reported outliers, indicating the accuracy of outlier detection. It is defined in Eq. 4:

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

- **Sensitivity** or recall quantifies the proportion of actual anomalies that are correctly identified, reflecting the comprehensiveness of the detection process. This is expressed in Eq. 5:

$$Sensibility = \frac{TP}{TP + FN} \tag{5}$$

- **Specificity** measures the ratio of correctly identified non-anomalous instances to the total number of true negatives, as shown in Eq. 6:

$$Specificity = \frac{TN}{TN + FP} \tag{6}$$

These evaluation metrics provide a rigorous framework for assessing the performance and effectiveness of anomaly detection algorithms (Creative Commons Attribution 4.0 International CC BY 4.0, 2020).

3. Methodology

The authors propose an advanced computational intelligence technique aimed at enhancing the identification of anomalies in electricity consumption, utilizing the defined thresholds outlined in Section 1. This suggested method offers a comprehensive approach applicable to various types of electrical usage, aiming to optimize accuracy while simultaneously reducing the FPR and FNR. Moreover, it enables the recognition of contextual anomalies within the electricity demand patterns. The proposed methodology encompasses five distinct stages, as illustrated in Fig. 2: (1) acquisition of historical data, (2) data processing, (3) creation of a dataset for the machine learning model, (4) model training and validation, and (5) presentation of the obtained results. For a visual representation, the proposed methodology is visually represented in Fig. 7 through a simplified block diagram.

3.1. Acquisition of historical data

In this study, electrical consumption data is studied for five buildings, selected for their diverse characteristics and representative nature of different electricity consumption patterns. These buildings are:

- **UPS_CENTRAL:** Electrical power associated with the connection of the power supply system at the Salesian Polytechnic University (UPS) in Cuenca, Ecuador. This building represents an educational institution in a developing country context.
- **UPS_CM:** Cornelio Merchán Building located in the Salesian Polytechnic University. This building offers insights into energy usage within a different structural and functional context in the same university campus.
- **UPV:** Building 5E of the Polytechnic University of Valencia, providing a perspective on electricity consumption in an academic setting in a developed country.
- **BRITHISH:** Electrical power of 20 residences in British Columbia, representing residential energy consumption in a North American context (Creative Commons Attribution 4.0 International CC BY 4.0, 2020).
- **SEAUX:** Electricity consumption data from a residential property situated in SEAUX, near Paris, France. This data offers a view of domestic energy usage in a European urban setting (Makonin, 2018).

The choice of these buildings provides a comprehensive range of data, encompassing different building types (academic, residential), geographical locations (Europe, North America, South America), and usage patterns. This diversity ensures that the developed model is robust and adaptable to various scenarios. The historical data used to develop the model for identifying anomalous usage amounts to a year (360 days) for each of the listed buildings, with time series data at intervals of 15 min. The varying locations of the installations mean that each building exhibits unique patterns of power usage, thereby enriching the dataset and enhancing the generalizability of the model.

3.2. Historical data processing

The proposed methodology aims to analyze and characterize the electricity consumption patterns of individual buildings. To accomplish this, the statistical approach of the SAECC technique is employed, leveraging historical data that encompasses both typical and atypical electricity usage. As depicted in Fig. 3, a whisker plot serves as an illustrative example, showcasing the normal electricity consumption trend for a specific day of the week. This graphical representation aids in identifying deviations from the expected pattern and contributes to a comprehensive understanding of electricity usage behavior.

Table 2
A range of distinct anomalous behavioral patterns identified in the context of daily energy consumption.

1. The analysis uncovers a standard consumption pattern falling within the anticipated range.	2. Potential registration errors are indicated by the presence of multiple samples exhibiting similar values.	3. Abnormal consumption is more prevalent during regular working hours.
4. There are no instances of extraordinary consumption recorded outside of business hours.	5. No unusual consumption observed during the morning hours outside of the typical work period.	6. Unusual consumption patterns are observed at night, potentially due to sustained charging.
7. Unusual consumption patterns are observed in the early morning hours.	8. No abnormal energy consumption observed during nighttime hours outside of the regular working period, suggesting the possibility of a connected charge from the previous day.	9. Anomalous consumption patterns are observed during nighttime hours, potentially attributed to continuous charging activities.
10. There is a substantial frequency of atypical data during the early morning hours outside of the typical working period.	11. There is a significant occurrence of abnormal data during nighttime hours.	12. Unusual data patterns are more prevalent during non-working hours, but daily energy consumption remains unchanged.
13. Unusual data patterns are more prevalent during non-working hours, but daily energy consumption remains unchanged.	14. There are limited occurrences of abnormal data within a day of energy consumption, and there are no significant fluctuations in the overall daily energy usage.	15. During working hours, abnormal consumption is observed more frequently, yet there are no significant variations in the overall daily energy consumption.
16. The electrical energy usage surpasses the anticipated pattern, suggesting enhanced efficiency in energy utilization throughout the entire day.	17. The electrical energy consumption falls below the predicted pattern, indicating a consistent trend of low power usage throughout the entire day.	18. Maximum power consumption exceeds the expected pattern, with additional consumption extending into peak hours.
19. The maximum power consumption deviates from the expected pattern, with reduced power usage observed even during peak hours.	20. Lowest power consumption exceeds the expected pattern, with additional consumption throughout the day.	21. Minimum power consumption falls below the expected pattern, indicating consistently low power consumption throughout the day.
22. Energy consumption exceeds the expected pattern, most likely due to a sustained charge or underlying issue.	23. Energy consumption is lower than expected, suggesting a potential connection issue or fault.	24. Maximum power consumption exceeds the expected pattern.
25. Maximum power consumption falls below the expected trend without a clear explanation.	26. Minimum power consumption exceeds the expected pattern, indicating a possible connection issue or fault.	27. Minimum power consumption falls below the expected pattern, suggesting a possible connection issue or fault.
28. Energy consumption exceeds the expected pattern, possibly due to increased work activity.	29. The lowest power consumption exceeds the expected pattern, indicating a potential connection of a charge or the presence of a fault.	30. The highest power consumption surpasses the anticipated pattern, potentially attributable to heightened work activity.
31. Maximum power consumption falls below the expected pattern, possibly due to decreased work activity.	32. Minimum power consumption exceeds the expected pattern, suggesting a possible connection issue or fault.	33. Minimum power consumption falls below the expected pattern, indicating a potential connection issue or fault.

3.2.1. Data normalization

This method introduces the use of a normalization of the electrical consumption data to the well-known **Z** values; these values enable the discrimination of the data that fall beyond a range set by the analyst. Eq. 7 can be used to calculate these values, which depend on changes in the mean from one sample to the next.

Table 3
Neural Network Parameters and Hyperparameters.

Parameters and Hyperparameters	
The count of neurons in the input layer	99
The count of neurons in the output layer	33
The count of hidden layers	1
number of neurons in the hidden layer	[100]
Dropout	0.5
activation function	'Relu'
training algorithm	'ADAM'
learning factor	0.01

$$X_z = \frac{X_t - \overline{X}}{\delta} \tag{7}$$

Where:

$\overline{X}$ represents the mean of the analyzed data.
δ represents the standard deviation of the analyzed data.
X_t represents the original value.
X_z represents the normalized value.

In this equation, the Z-score, X_z is calculated by subtracting the mean value, $\overline{X}$, from the original data point, X_t, and then dividing the result by the standard deviation, δ. This process transforms the original data values into a standardized form where the mean is 0 and the standard deviation is 1. Normalizing the data in this way allows for the comparison of data points from different distributions or scales, facilitating the identification of outliers and unusual patterns in the electrical consumption data.

Fig. 4 shows the same example as Fig. 1, but this time the electrical consumption is examined in the **Z** domain; the electrical consumption pattern is now shown as a symbolized interval in **Z** values that is defined by a range between a maximum **Z** value and a minimum **Z** value rather than a box-plot diagram. This graph facilitates the comprehension of the idea that a day of consumption is deemed anomalous if any of its numbers fall outside the range.

3.2.2. Cataloging changes in electrical consumption

This research not only determines whether an anomaly has happened, but also suggests a potential cause. To identify the most common abnormal behaviors in daily energy consumption, various factors were taken into account, including the proportion of atypical values, the timing of occurrence, the duration, subsequent measurements with identical values, and, importantly, the patterns exhibited by the maximum and minimum power. These criteria allow the identification of 33 different types of anomalies-causing alterations, which are shown in Table 2.

3.3. Machine learning data set creation

Each building being analyzed has its own data processing, as seen in Fig. 5:

1. The first step is to process all the historical data from each of the databases that have been analyzed in order to produce a DLP, transform the input data to Z Domain, and catalog the different types of anomalous consumption using Table 2.
2. Following the previous procedure, the day of analysis and the kinds of anomalies they reveal are both recorded in a database. Next, all the databases produced by the various buildings are combined into a single database.
3. Using the Z-domain DLPs as model input and the cataloging of each DLP as model output, an automated learning model is trained using the final database. The empirical rule of 80/20 is used to split training data into 20% for testing and 80% for training.

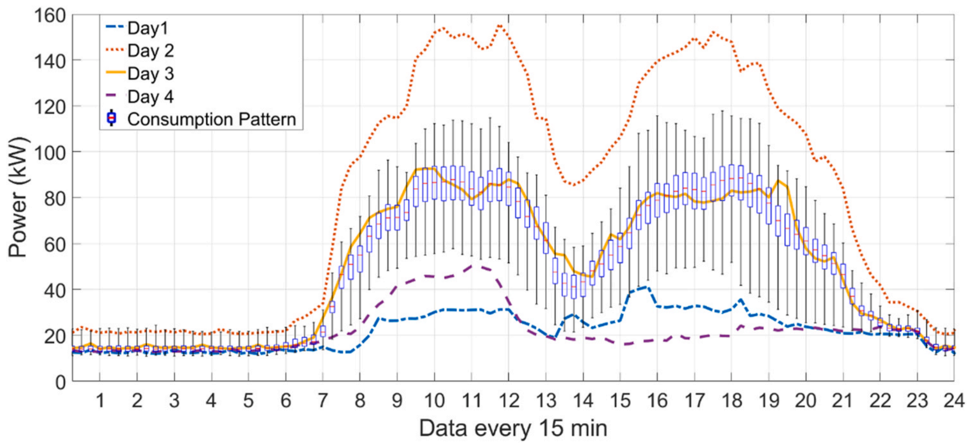

Fig. 8. Consumption pattern and analyzed days 1–4 daily load profiles.

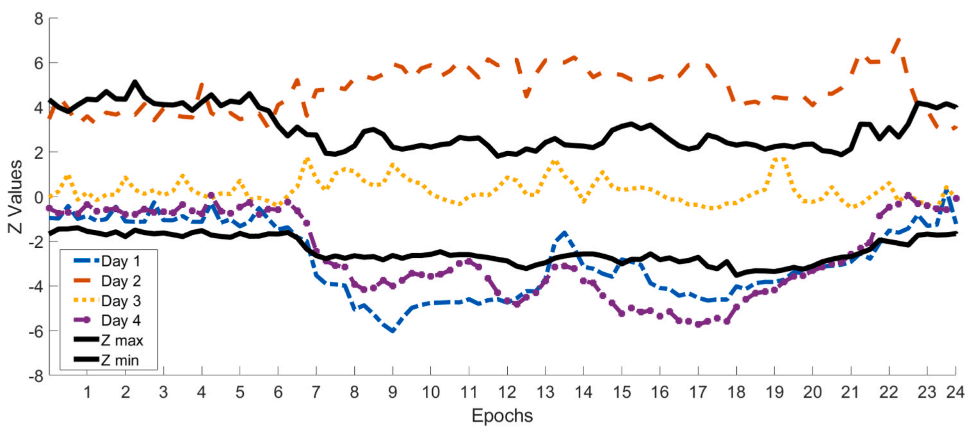

Fig. 9. Consumption pattern and analyzed days 1–4 daily load profiles in Z domain.

3.4. Model training and Validation

The learning model aims to catalog the various types of anomalies shown in Table 2, where:

- The system inputs correspond to electrical power data with a 15 min time step of a full day (24 h = 96 intervals) in the "Z" domain, the value of the maximum, average, and minimum power, giving a total of 99 inputs.
- The system's output matches to a list of 33 different forms of abnormal behavior.

The next issue is the classification, where there are 99 entries relating to a day's worth of electricity consumption characteristics and 33 outputs related to various forms of outliers in Table 2. This work proposes an artificial neural network to address this classification challenge, whose design is shown in Fig. 6.

The training algorithm selected is ADAM (A Method for Stochastic Optimization) (Kingma and Ba, 2015), and the initial parameters are known to be the number of inputs equal to 99 and the number of outputs equal to 33. The "DropOut" strategy is then applied to try to prevent overfitting, in order to avoid issues with improper initialization of parameters and to enable an improvement in the convergence of the neural network.

The next step is to define the hyperparameters of the neural network:

- The learning factor.
- The quantity of hidden layers and their corresponding number of neurons.
- The selection of activation functions for the hidden layers.

This work uses the hyperparameter optimization methodology known as gradient-powered regression trees to create these hyperparameters. This technique boosts a model incrementally by computing the gradient with respect to the hyperparameters and optimizing the values. The neural network's parameters and hyperparameters are shown in Table 3.

4. Results

4.1. Model training

The training of neural network is carried out with the parameters and hyperparameters defined in the previous section, for which the training dataset is subdivided into (80−20) for training and validation during each training epoch using a K5 fold validation. Once this is finished, it is possible to obtain the Accuracy and loss curves during each training epoch, these curves are shown in Fig. 7, respectively. It can be seen that the accuracy both in training and in validation reaches a value of 95%. On the other hand, the loss function reaches a value less than 0.11 both in its training and in its validation.

4.2. Model evaluation

The method uses 20% of the split data for testing in the final stage,

Table 4
Conclusions of electrical consumption of 4 analyzed days.

Day Analysis	Day of the Week	Is Truly Anomalous?	Is Anomalous with Proposed Method?	Conclusions of methodology
1	Wednesday	Yes	Yes	• No abnormal energy consumption observed during nighttime hours outside of the regular working period, suggesting the possibility of a connected charge from the previous day. 8 • The lowest power consumption exceeds the expected pattern, indicating a potential connection of a charge or the presence of a fault. 29 • Maximum power consumption falls below the expected pattern, possibly due to decreased work activity.31
2	Wednesday	Yes	Yes	• Energy consumption exceeds the expected pattern, most likely due to a sustained charge or underlying issue.22 • Maximum power consumption exceeds the expected pattern.24 • Minimum power consumption exceeds the expected pattern, indicating a possible connection issue or fault. 26
3	Wednesday	No	No	• The analysis uncovers a standard consumption pattern falling within the anticipated range. 1
4	Wednesday	Yes	Yes	• The lowest power consumption exceeds the expected pattern, indicating a potential connection of a charge or the presence of a fault. 29 • Maximum power consumption falls below the expected pattern, possibly due to decreased work activity.31

which assesses data that the neural network has not seen in training or validation, yielding an accuracy score of 95.54%. It allows us to confirm that the model was trained properly and without overfitting.

We use the **UPS_CENTRAL** building as an analysis example while highlighting the fact that the same outcomes apply to the other analysis buildings. After identifying the features and going through the grouping procedure, as was discussed in Section 3.2, the final consumption pattern is created for each sort of day of the week, in this case a "Wednesday." Fig. 8 displays the consumption pattern (box plots) and the consumption of the investigated days (96 power values). In order to present the information clearly, the data is also translated to the Z domain as indicated in Section 3.2.1 (Fig. 9). Days 1, 2, and 4 were examined in relation to days where anomalous values were found. The third day under study corresponds to a DLP with normal consumption and no outliers. For each DLP that has been examined, Table 6 displays the inference of the kinds of outliers that this approach produces.

The application of the proposed method to the UPS_CENTRAL building has yielded promising results in analyzing electrical consumption and detecting anomalies. By utilizing an approach based on the creation of consumption patterns for different days of the week, a deeper understanding of both normal and abnormal energy consumption behaviors has been achieved. Moreover, the translation of data into the Z domain has facilitated a clearer and more comparative representation of daily load profiles. Through a detailed analysis of the four selected days, valuable insights regarding the building's electrical consumption have been uncovered. Patterns indicative of anomalies have been identified on specific days, suggesting potential issues such as inappropriate load connections or system failures. These observations provide strong support for the effectiveness of the proposed method in detecting atypical events and offer relevant information for enhancing energy efficiency.

5. Discussion

The anomaly detection analysis for 20% of the evaluation data is shown in Table 4 for each of the analyzed buildings. The details include the count of DLPs classified as typical, anomalous, or genuinely anomalous (based on expert assessment), the count of anomalous DLPs correctly identified as anomalous (true positives), anomalous DLPs mistakenly labeled as non-anomalous (false negatives), the false positive rate (FPR), false negative rate (FNR), and precision. The same analysis as in the preceding paragraph is also shown in Table 4, but using the SAEEC statistical method, in order to compare the outcomes of the suggested method.

In addition, using the Eq. 8, it is possible to quantify the relative improvement (RI) of precision, FPR, and FNR of the proposed method compared to an SAEEC method. These results can be seen in Table 5. The findings indicate that employing the suggested approach enhances the identification of abnormal DLPs by extrapolating the electricity consumption patterns from a diverse building database. As a result, it enables the detection of challenging contextual anomalies that would otherwise be difficult to identify.

$$Relative_{Improvement} = \frac{|Current_{Value} \quad - New_{Value}|}{Current_{Value}} \tag{8}$$

Upon comparing the results of the proposed method with the SAEEC statistical method in Table 4, it is evident that the proposed method exhibits significant improvements in detecting anomalous DLPs. Specifically, for the UPS_CENTRAL building, the proposed method shows a lower FNR and a higher TP count compared to the SAEEC method, indicating its enhanced ability to accurately identify anomalous DLPs in this specific building. Similarly, the proposed method demonstrates improvements in terms of FPR, FNR, and precision across the other analyzed buildings when compared to the SAEEC method.

Furthermore, Table 5 presents the relative improvement (RI) values of precision, FPR, and FNR achieved by the proposed method when

Table 5
Analyzing and contrasting the technique for anomaly detection.

		TrulyAnomalousDLP	DLP reported as Typical	DLP reported as Anomalous	Anomalous DLP Reported as Anomalous	Anomalous DLP Reported as not Anomalous	FPR%	FNR%	Precision %
SAECC	UPS_CENTRAL	127	59	136	103	33	48,529	18,898	75,735
	UPS_CM	122	60	136	109	27	36,486	10,656	80,147
	UPV	104	83	112	85	27	29,670	18,269	75,893
	BRITHISH	536	372	602	457	145	33,105	14,739	75,914
	SEAUX	183	181	208	158	50	24,272	13,661	75,962
PROPOSED	UPS_CENTRAL	127	64	131	120	11	16,176	5512	91,603
	UPS_CM	122	67	129	111	18	24,324	9016	86,047
	UPV	104	88	107	93	14	15,385	10,577	86,916
	BRITHISH	536	384	590	517	73	16,667	3545	87,627
	SEAUX	183	186	203	160	43	20,874	12,568	78,818

Table 6
Relative Improvement of the Proposed method in each building analyzed.

	Relative Improvement		
	FPR	FNR	Precision
UPS_CENTRAL	66,67%	70,83%	20,95%
UPS_CM	33,33%	15,38%	7,36%
UPV	48,15%	42,11%	14,52%
BRITHISH	49,66%	75,95%	15,43%
SEAUX	14,00%	8,00%	3,76%
Average	**42,36%**	**42,45%**	**12,41%**

compared to the SAEEC method. The results reveal substantial enhancements in detecting anomalous DLPs across all the analyzed buildings. The average relative improvement is 42.36% for FPR, 42.45% for FNR, and 12.41% for precision. These findings underscore the superior performance of the proposed method in generalizing energy consumption behavior across diverse buildings, thereby enabling the identification of contextual anomalies that are challenging to detect using the SAEEC method.

6. Conclusion

The research presents a novel computational intelligence method for detecting and evaluating changes in the electrical consumption patterns of buildings. This method offers several benefits, including effective energy management, cost reduction, anomaly detection, fault identification, and time-saving for technicians and facility managers. It enhances the dependability of surveillance operations in building systems.

The study employed statistical analysis of confidence intervals and data normalization towards the "Z" value to categorize and classify daily consumption patterns in various buildings. By utilizing a machine learning model, the method achieved a high precision rate of 95% during training, validation, and testing stages.

Comparison with a purely statistical approach revealed a significant improvement in detecting atypical electrical consumption data across all buildings analyzed. The proposed method demonstrated a 12.41% increase in precision and a substantial decrease of 42.3% and 42.45% in false positive and false negative rates, respectively. These results affirm the method's superior accuracy and reliability in anomaly detection.

Furthermore, the proposed method exhibits the capability to analyze and deduce potential causes behind anomalous consumption profiles in real-time. This real-time analysis enables prompt identification of energy consumption issues, expense reduction, and overall improvement in energy efficiency. It also facilitates the evaluation of the effectiveness of energy-saving programs implemented in buildings.

While acknowledging the potential limitations of the method, such as specific building factors and seasonal changes impacting consumption patterns' representativeness, the obtained results validate the utility and efficacy of the proposed method. To address these limitations, future research could incorporate more dynamic modeling techniques that account for seasonal variations more effectively. Exploring adaptive algorithms that can adjust to changing conditions and integrating a broader range of environmental and operational data could potentially enhance the method's accuracy and representativeness.

Additionally, there is an opportunity to refine the detection of false positives and negatives by employing more advanced machine learning algorithms and more extensive datasets. These enhancements would allow the system to better understand and adapt to the unique energy usage patterns of different buildings and seasons.

In conclusion, the proposed method offers an automated approach to assess and deduce reasons behind anomalous energy consumption profiles. It not only assists in reducing energy expenditures and consumption but also aids in promptly identifying anomalous usage and problems. Additionally, it contributes to improving energy efficiency and evaluating the effectiveness of energy-saving programs in buildings. Moving forward, the method will evolve to more precisely address the complexities of energy consumption patterns, adapting to seasonal variations and other specific challenges, thereby pushing the boundaries of smart energy management.

Ethical Approval

This study did not involve human or animal experimentation. All procedures performed in the study were in accordance with institutional and national ethical standards, as well as the 1975 Helsinki Declaration and its later amendments or comparable ethical standards.

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This study did not receive external funding.

Author contributions

Santiago Luna: Principal investigator and lead author of the study. Responsible for the conception and design of the computational intelligence method, leading the data collection, analysis, and interpretation. Primary author in manuscript writing and coordination of co-authors' contributions. Xavier Serrano-Guerrero: Contributed to the review of the methodology and practical application of the proposed method, providing critical and constructive perspectives for the study's refinement. Guillermo Escrivá-Escrivá: Provided expert input in the study area, advising on technical and theoretical aspects. Essential contribution to the writing and structuring of the text, ensuring academic precision and relevance. Mauren Abreu de Souza: Assisted in the final manuscript revision, providing corrections and suggestions to enhance textual coherence and quality.

CRediT authorship contribution statement

Luna Romero Santiago Felipe: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Serrano-Guerrero Xavier:** Supervision, Validation, Writing – review & editing. **Abreu de Souza Mauren:** Writing – review & editing. **Escrivá-Escriva Guillermo:** Supervision, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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